

IBRAHIM AREMU MIKAIL

Healthcare AI Engineer · Medical Student (MBBS) · Building Agentic Systems at the Intersection of Medicine and Machine Intelligence

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PROFESSIONAL SUMMARY

Healthcare AI Engineer and medical student with an MSc in Public Health and a BSc in Human Anatomy. I design and deploy end-to-end agentic AI systems for clinical use including LLM-powered tools, RAG pipelines, ML inference pipelines with MLOps infrastructure, and full-stack clinical decision support applications. My dual identity as both clinician in training and AI engineer means I understand the disease and the data pipeline that processes it. Current focus areas include explainable AI for oncology, vascular disease modeling, health equity in algorithmic systems, and agentic workflows for low-resource clinical settings.

EXPERIENCE

Clinical Research Assistant · Federal Medical Centre, Abeokuta — Dept. of Pain & Palliative Medicine2024 – Present

- Managed extraction of longitudinal patient data from EMR systems via APIs, ensuring data integrity for clinical research and outcome analysis.
- Collaborated with clinical teams to translate raw patient records into actionable statistical insights, informing departmental decision-making on pain management protocols.
- Conducted exploratory and predictive analysis on neuropathic pain trends in palliative care patients; developed ML models for pain control outcome prediction.
- Developed and presented dashboards to the Head of Department summarizing clinical findings and recommending data-driven interventions.

Healthcare Analyst · Unified Mentor — Remote2025 – 2026

- Designed interactive clinical dashboards to visualize patient data trends, ensuring high accuracy in stakeholder reporting across ER operations.
- Performed exploratory data analysis to identify operational bottlenecks in emergency room workflows, proposing data-driven resource optimization strategies.

PROJECTS

AuxiChron — Digital Health Twin Flagship

Agentic clinical OS for chronic care management. Built fullstack with AI-powered triage, longitudinal patient monitoring, and a provider command center. Architected a RAG pipeline grounded in the Nigerian Standard Treatment Guidelines, ensuring transparent, clinically safe outputs. Deployed as a production application.

Stack: LangChain · LangGraph · RAG · FastAPI · Docker · React | [Live Demo](#)

OncoSight-SSA — Multi-Agent XAI for Cancer Care in Sub-Saharan Africa In Progress

First integrated pipeline combining cancer risk screening (breast & cervical), oncology facility mapping, referral recommendation, and patient compliance prediction for low-resource Sub-Saharan African settings. Each of four coordinated agents is paired with SHAP, LIME, and counterfactual explainability. Trained on Kenya DHS, Kenya Health Facilities, and WHO/World Bank datasets.

Stack: LangGraph · XGBoost · SHAP · LIME · Alibi · Groq · GeoPandas · FastAPI · Docker | [Documentation](#)

BCT LLM Agent — Nigerian User Modeling & Recommendation System

Agentic recommendation system built for the DSN × Bluechip Technologies Hackathon 3.0. Implements two tasks: user modeling (simulates how a specific Nigerian user reviews an unseen item with cultural nuance) and personalized recommendation with multi-turn memory and cold-start handling. Demonstrates the agentic RAG architecture applied to contextual AI in low-resource settings.

Stack: LangChain · LlamaIndex · ChromaDB · Groq · FastAPI · Docker · HuggingFace Spaces | [Live Demo](#)

Life Expectancy Analysis — Global Longevity Prediction

Machine learning approach to predicting global life expectancy from socio-economic and public health drivers. Multi-model comparison with feature importance analysis, deployed as an interactive Streamlit application.

Stack: Python · Scikit-learn · Streamlit | [Live App](#)

Hospital Readmission Drivers & Root Cause Analysis

End-to-end analysis of hospital readmission drivers across hospital, community, and daycare services. Achieved a validated 10% readmission rate against a 13.9% sector benchmark. Developed 20 KPIs and 3 thematic Power BI dashboards for leadership decision-making; included equity analysis across demographic and socioeconomic variables.

Stack: Power BI · DAX · SQL | [Documentation](#)

TECHNICAL SKILLS

AI & Agentic Systems: LangChain, LangGraph, RAG Pipelines, Agentic Workflows, LLM Engineering, Prompt Engineering, Groq API

Machine Learning: Python (Scikit-learn, XGBoost, Pandas, NumPy), Predictive Modeling, ML Pipelines, NLP in Healthcare

Explainability: SHAP, LIME, Counterfactual Explanations (Alibi)

Backend & Deployment: FastAPI, Docker, HuggingFace Spaces, Streamlit, AWS Cloud

Frontend: React, React Native, Full-stack Development

Data & Biostatistics: Advanced SQL, SPSS / Stata / SAS, GLMs, Multilevel Modeling, Risk Adjustment (HCC), GeoPandas

Research & Epidemiology: Precision Oncology, Study Design (RCT/Cohort), ICD-10 Coding, Systematic Reviews, Meta-Analysis

Clinical & Governance: EMR Architecture, HL7/HIPAA, Clinical Data Standards, AI Ethics & Fairness, Public Health Policy

Visualization: Power BI, Tableau, Matplotlib / Seaborn

DevOps & Version Control: Git, GitHub, Docker

EDUCATION

MBBS — Bachelor of Medicine & Bachelor of Surgery · EKOUNIMED 2023 – Present

MSc Public Health · National Open University of Nigeria

BSc Human Anatomy · Olabisi Onabanjo University 2023

– First Class — GPA: 4.21 / 5.0

CERTIFICATIONS

- **Healthcare Data Science & AI in Healthcare** — Digital Health Africa
- **Google Data Analytics Professional Certificate** — Coursera · 2025
- **Data Scientist Professional Certificate** — DataCamp
- **Medical Neuroscience (Graduate-Level)** — Duke University via Coursera · 2025

REFERENCE

Dr. Ayilara S.O. — Consultant Family Physician

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